
A TEMPLATE FOR THE *arxiv* STYLE

A PREPRINT

 **Julius Fenn**

Cluster of Excellence *livMatS* @ FIT–Freiburg Center for Interactive Materials and Bioinspired Technologies,
University of Freiburg
Georges-Köhler-Allee 105, 79110 Freiburg, Germany
Institute for Psychology, University of Freiburg
79085 Freiburg im Breisgau, Germany
`julius.fenn@psychologie.uni-freiburg.de`

 **AAA BBB***

Department of XXX
XXX University
Freiburg, XXX
`XX@YYY`

March 18, 2026

ABSTRACT

Keywords First keyword · Second keyword · More

1 Introduction

This section provides a research-grade synthesis over the PDF corpus in `./context/`. All cited works are included in `output/references.bib`.

1.1 Phase 1: Rigorous term definitions and distinctions

Across this corpus, terminology is not fully standardised: the same observed phenomenon is variously framed as bots, fraud, misrepresentation, or imposter participation depending on discipline and research setting [Husted et al., 2025, Pinzón et al., 2024, Xu et al., 2022]. The corpus provides limited direct treatment of legal/policy definitions; where such framing is needed, it should be supported by additional sources not contained in `./context/`.

1.1.1 Imposter

Key definition (canonical). An *imposter participant* is a respondent who participates while misrepresenting identity or eligibility, either fully or partially, in a way that threatens data integrity [Husted et al., 2025, Ridge et al., 2023, Sharma et al., 2024].

Multiple literal definitions/usages in this corpus.

1. **Qualitative research context:** Imposter participation is conceptualised as misrepresentation of identity, either in full or partially [Husted et al., 2025].

*Use footnote for providing further information about author (webpage, alternative address)—*not* for acknowledging funding agencies.

2. **Online qualitative recruitment context:** Researchers describe “imposter participants” as an emerging threat to data integrity in online qualitative studies, with indicators such as inconsistent recruitment-source claims and suspicious contact patterns [Ridge et al., 2023, Santinele Martino et al., 2024, Sharma et al., 2024].

Differentiation from the other terms.

1. **Imposter vs. fraud:** In this corpus, “imposter” is sometimes preferred over “fraudulent” to avoid inferring criminal intent when the core issue is ineligibility or identity misrepresentation [Husted et al., 2025].
2. **Imposter vs. bot:** Imposters can be human actors (or human-operated accounts) and need not be automated; bots are defined as automated programs/applications simulating human actions [Xu et al., 2022, Pinzón et al., 2024].
3. **Imposter vs. LLM pollution:** Imposter participation concerns (mis)representation of identity/eligibility; LLM pollution concerns the mediation/delegation of responses to LLMs even when the participant identity may be genuine [Husted et al., 2025, Rilla et al., 2025].
4. **Imposter vs. AI agent:** Imposters are a participation role/category; AI agents are system capabilities (autonomy/tool-use) that may or may not be used to impersonate or misrepresent [Agley, 2025, Westwood, 2025].

Operational interpretation (recognition in practice). Observable properties include inconsistencies with eligibility claims, inconsistent recruitment-path narratives, refusal to verify identity where verification is required for the study, repeated patterns across contact attempts, and other indicators suggesting misrepresentation rather than ordinary respondent error [Ridge et al., 2023, Husted et al., 2025, Sharma et al., 2024].

Boundary cases. A genuine participant using a pseudonym for privacy may be “imposter” only if the misrepresentation invalidates eligibility/analysis assumptions; a human who is eligible but uses automation to complete tasks is better characterised as automation/LLM delegation than “imposter” unless identity/eligibility is misrepresented [Husted et al., 2025, Rilla et al., 2025].

1.1.2 Fraud

Key definition (canonical). *Survey fraud* is the intentional production of inauthentic, low-quality, or misrepresentative submissions (often to obtain incentives) that undermine validity, including both human fraudsters and automated bots [Pinzón et al., 2024, Pozzar et al., 2020, Maline and Polonijo, 2025].

Multiple literal definitions/usages in this corpus.

1. **Operational fraud labeling (health survey case study):** Indicators attributable to respondent error are labelled *suspicious*, while those suggesting misrepresentation are labelled *fraudulent* [Pozzar et al., 2020].
2. **AI-powered fraud framing:** Fraud is presented as an integrity threat driven by both “AI bots” and “human fraudsters,” with emphasis on financial incentives and adaptive evasion [Pinzón et al., 2024].
3. **Platform recruitment setting (MTurk):** A substantial fraction of submissions can be characterised as fraudulent under multi-indicator screening protocols (e.g., duplicate demographics, repeated coordinates, nonsensical open-ended responses) [Maline and Polonijo, 2025].

Differentiation from the other terms.

1. **Fraud vs. imposter:** Fraud is intent- and harm-oriented (e.g., gaming incentives, corrupting data); “imposter” is often used to describe ineligibility/misrepresentation without necessarily asserting criminal intent [Husted et al., 2025, Pinzón et al., 2024].
2. **Fraud vs. bot:** Bots are a means (automation) that can be used for fraud, but fraud can be human-only; conversely, automation used for benign purposes is not necessarily fraud [Xu et al., 2022, Pinzón et al., 2024].
3. **Fraud vs. LLM pollution:** LLM pollution can occur without adversarial intent (participants seeking help/translation), whereas fraud typically assumes strategic intent to mislead or extract incentives [Rilla et al., 2025, Pinzón et al., 2024].
4. **Fraud vs. AI agent:** Agents may enable fraud (e.g., autonomous completion), but agents can also be non-adversarial tools; fraud is a behaviour/outcome category [Agley, 2025, Westwood, 2025].

Operational interpretation. In applied protocols, fraud is recognised via multi-indicator evidence: inconsistent verifiable items, duplicate or templated open-ended text, suspicious network/location signals, anomalous timing/bursts, and patterns consistent with automation [Pozzar et al., 2020, Maline and Polonijo, 2025, Pinzón et al., 2024, Bonett et al., 2024].

Boundary cases. A respondent who is inattentive but not deceptive may be low-quality but not fraudulent; a participant using an LLM for wording/translation can degrade measurement without being fraudulent (LLM pollution) [Pozzar et al., 2020, Rilla et al., 2025].

1.1.3 LLM pollution

Key definition (canonical). *LLM pollution* denotes threats to online behavioural research validity arising when participants use LLMs for advice, translation, or task delegation, such that outputs are shaped by (or produced by) an LLM rather than the intended human cognitive process [Rilla et al., 2025].

Multiple literal definitions/usages in this corpus. [Rilla et al., 2025] explicitly distinguishes three variants: (i) *Partial LLM Mediation* (selective use of LLMs), (ii) *Full LLM Delegation* (agentic LLMs complete studies with little/no human oversight), and (iii) *LLM Spillover* (humans change behaviour anticipating LLM presence).

Differentiation from the other terms.

1. **LLM pollution vs. bot:** LLM pollution includes both human-mediated and fully delegated use; it is not identical to “survey bots” as automated random form fillers [Rilla et al., 2025, Xu et al., 2022].
2. **LLM pollution vs. fraud:** LLM pollution may be non-adversarial and motivated by assistance; fraud is typically adversarial or incentive-driven misrepresentation [Rilla et al., 2025, Pinzón et al., 2024].
3. **LLM pollution vs. AI agent:** Full LLM Delegation is a specific agentic mechanism within LLM pollution; not all AI agents imply pollution (e.g., agents used by researchers for tooling) [Rilla et al., 2025, Agley, 2025].
4. **LLM pollution vs. imposter:** LLM pollution concerns how responses are produced; imposter concerns identity/eligibility misrepresentation [Rilla et al., 2025, Husted et al., 2025].

Operational interpretation. Observable properties include copy/paste traces, response-generation patterns consistent with LLM use, and task-level evidence of mediation/delegation even when standard bot/fraud checks pass [Rilla et al., 2025, Claassen et al., 2026, Lebrun et al., 2024a].

Boundary cases. Translation help for non-native speakers can be partial mediation; an “autonomous synthetic respondent” completing surveys constitutes full delegation; behavioural spillover can occur even without any LLMs present [Rilla et al., 2025, Westwood, 2025].

1.1.4 AI agent

Key definition (canonical). In this corpus, an *AI agent* is an AI system that can pursue a goal by selecting and executing actions (potentially using tools such as a web browser), with some degree of autonomy beyond single-turn text generation [Agley, 2025, Westwood, 2025, Rilla et al., 2025].

Multiple literal definitions/usages in this corpus.

1. **Computer-using agent (CUA) framing:** [Agley, 2025] describes OpenAI’s “Operator” as an agent that can use its own browser to interact with webpages by typing, clicking, and scrolling.
2. **Autonomous synthetic respondent framing:** [Westwood, 2025] reports an autonomous synthetic respondent/AI agent that evades standard quality checks and produces coherent, internally consistent survey data.
3. **Agentic LLM delegation framing:** [Rilla et al., 2025] frames full delegation as agentic LLMs completing studies with little to no human oversight.

Differentiation from the other terms.

1. **AI agent vs. bot:** “Bot” in this corpus often denotes automated form fillers or automated applications simulating human actions; an AI agent is a broader capability class that can include tool use and autonomy

(and may be much more capable than random-response bots) [Xu et al., 2022, Pinzón et al., 2024, Agley, 2025, Westwood, 2025].

2. **AI agent vs. LLM pollution:** Agents can cause LLM pollution when delegated to complete studies, but AI agents also exist outside research participation (e.g., task automation) [Rilla et al., 2025, Agley, 2025].
3. **AI agent vs. fraud/imposter:** Fraud/imposter are intent/role categories; an AI agent is a mechanism that can be used benignly or adversarially [Husted et al., 2025, Pinzón et al., 2024].

Operational interpretation. Observable properties include tool-use traces (browser interaction patterns), coherent long-horizon behaviour (persona memory/consistency), and the ability to complete multi-step tasks without continuous human input [Agley, 2025, Westwood, 2025].

Boundary cases. A human using a single LLM prompt for wording is not necessarily an agent; a computer-using system that navigates webpages and executes actions qualifies as an agentic system; a human “puppeteer” controlling many accounts may not be an AI agent even if automation assists the workflow [Agley, 2025, Westwood, 2025, Wang et al., 2025].

1.1.5 Bot

Key definition (canonical). A *bot* is an automated program/application that simulates human actions to submit responses or operate accounts, including automated form-filling in surveys and automated behaviours on platforms [Xu et al., 2022, Pinzón et al., 2024, Storozuk et al., 2020].

Multiple literal definitions/usages in this corpus.

1. **Survey research context:** “Survey bots” are “automated form fillers” that fill out web-based surveys with random responses [Xu et al., 2022].
2. **Platform-integrity context (survey fraud literature):** Bots are “automated software applications simulating human actions” that can be programmed to automatically respond and generate invalid data, often for financial gain [Pinzón et al., 2024].
3. **Web-survey experimental context:** Bots can be rule-based or AI-based and may successfully pass common countermeasures such as CAPTCHAs and honeypots [Höhne et al., 2025].

Differentiation from the other terms.

1. **Bot vs. fraud:** Bots are a means; fraud is the adversarial behaviour/outcome. Bots can be used for fraud, but fraud can also be human-only [Pinzón et al., 2024, Pozzar et al., 2020].
2. **Bot vs. AI agent:** AI agents are a broader, more capable subset when they exhibit autonomy and tool use; many “survey bots” are simpler automated scripts [Xu et al., 2022, Agley, 2025, Westwood, 2025].
3. **Bot vs. imposter:** Imposter participation can be human misrepresentation without automation; conversely, bots can be deployed without impersonating a specific identity [Husted et al., 2025, Xu et al., 2022].
4. **Bot vs. LLM pollution:** LLM pollution includes mediated human participation and agentic delegation; bots may be random-response automation and not involve LLMs [Rilla et al., 2025, Xu et al., 2022].

Operational interpretation. Observable properties include automation-like timing and interaction traces, anomalous network/location patterns, repeated/templated content, and failure modes where common bot checks are insufficient against more capable bots [Höhne et al., 2025, Bonett et al., 2024, Pinzón et al., 2024].

Boundary cases. “Bots” in this corpus also overlap with human-driven coordinated misuse (e.g., multiple accounts or misrepresentation) that may not be fully automated; some papers argue that what researchers call “bots” may include broader categories of inauthentic participation [Jaffe et al., 2026, Wang et al., 2025].

1.2 Phase 2: Identified AI agents in literature and practice (within this corpus)

Table 1 lists concrete agentic systems discussed in the corpus.

Table 1: AI agents identified in the ./context/ corpus.

Agent type	Representative system/family	Org.	Autonomy/type
Computer-using agent (CUA)	“Operator” (browser-using agent)	OpenAI (as described)	Tool-using; br
Autonomous synthetic respondent	Synthetic survey respondent/agent	Research prototype	Autonomous;
Agentic LLM delegation (category)	Full LLM Delegation	Category (multiple possible systems)	Autonomous;

1.3 Definitions: fraud, imposter participants, bots, and agents

Across online studies, the labels *fraud*, *imposters*, *bots*, and (*AI*) *agents* are often used interchangeably in informal discussion, but the literature typically distinguishes them by (a) whether the behavior is intentional deception versus error and (b) whether the response generation is automated versus human.

Fraud. In practice, *fraudulent* responding is usually treated as intentional misrepresentation that undermines sample validity and data integrity (e.g., to obtain incentives), and it is commonly operationalized via indicator-based screening. For instance, Pozzar et al. [2020] explicitly labeled indicators attributable to respondent error as *suspicious* while reserving *fraudulent* for patterns suggesting misrepresentation. Relatedly, web-based data integrity work often defines fraud to include automated and duplicative submissions (e.g., “fraudulent entries from automated bots and duplicative submissions”; Bonett et al. 2024).

Imposter participants. *Imposter participants* are typically described as individuals who present as eligible/real participants while misrepresenting identity, eligibility, or uniqueness (e.g., enrolling multiple times or participating from outside the target population), posing a direct threat to the credibility of online qualitative data [Ridge et al., 2023, Sharma et al., 2024].

Bots. *Bots* are commonly defined as automated software that performs tasks without a human respondent, including “automated form fillers” that complete web surveys [Griffin et al., 2022, Xu et al., 2022]. In remote digital trials, this is sometimes operationalized more narrowly as “automated responses generated in clusters,” distinguished from manual deception [Loebenberg et al., 2023].

Agents (AI agents). Recent work distinguishes *agentic* automation from earlier bot threats by emphasizing that vision-enabled AI agents can autonomously complete surveys through the same rendered interface humans see (e.g., by taking screenshots, interpreting questions, and submitting responses), which can make many traditional quality checks less diagnostic [M. Affonso, 2025]. In this framing, “bots” can be seen as a broad class of automated responders, whereas “AI agents” denote more general-purpose, perception-and-action systems capable of completing diverse study tasks with minimal or no human oversight.

To keep terms analytically separable, this manuscript uses *fraud* as the umbrella category (intentional misrepresentation), with *imposters* denoting predominantly human identity/eligibility deception and *bots/agents* denoting automated response generation. We treat *LLM pollution* as a related but not identical mechanism: it can occur via partial mediation (human participants using LLMs as assistance) or full delegation (agentic systems completing tasks), and it may also induce second-order spillover effects in human behavior [Rilla et al., 2025].

2 Percentages of AI pollution

2.1 on Mturk, Prolific

We synthesize empirical prevalence estimates for three partially overlapping threats to online survey data integrity on Amazon Mechanical Turk (MTurk) and Prolific. We define *strict fraud/bot* as responses classified as fraudulent or otherwise non-authentic based on clear evidence of misconduct (e.g., re-entry, duplication, implausible traces) or as accounts inferred to be controlled by a single actor (e.g., “puppet” accounts). We define *broad invalid data* as responses failing study-specific validity or “high-quality” criteria (e.g., attention and instruction checks, timing thresholds, open-ended response screening, internal consistency, and IP/geolocation uniqueness), acknowledging that such criteria bundle heterogeneous failure modes. We define *AI/LLM contamination* as the use of large language models (LLMs) to generate or materially mediate responses; in this corpus, prevalence is operationalized either via behavioral traces plus synthetic-text classification [Veselovsky et al., 2023] or via detector-based and signature-based estimates of AI-generated open-ended survey text [Zhang et al., 2025].

Across the available numeric estimates, Prolific tends to yield a larger share of “high-quality” respondents (and thus a lower implied invalid rate) than MTurk in cross-platform comparisons [Douglas et al., 2023, OConnell et al., 2025]. However, strict fraud/bot prevalence estimates are concentrated on MTurk, and no Prolific-specific strict fraud/bot prevalence estimate was identified in the loaded PDFs.

For MTurk strict fraud/bot, Maline and Polonijo report that 143/221 (64.71%) responses were classified as fraudulent in a November 2023 survey experiment, and also report a broader estimated range of 65–84% fraudulent responses using multiple fraud signals [Maline and Polonijo, 2025]. Wang et al. report that 33–56.4% of accounts in two MTurk studies are “likely puppets” (with study-level point estimates of 35.1% and 56.5%) based on behavioral detection of potential puppet accounts [Wang et al., 2025]. For broad invalid data, estimates vary sharply with screening definitions: Webb and Tangney report 14/529 (2.6%) “valid” respondents (implying 97.4% invalid by their criteria) [Webb and

Tangney, 2024]; Douglas et al. report 26.40% “high-quality” on MTurk (implying 73.60% not high-quality) and 67.94% “high-quality” on Prolific (implying 32.06% not high-quality) under their composite definition [Douglas et al., 2023]; and O’Connell et al. report that 9.89% of MTurk participants and 43.34% of Prolific participants produced “high-quality” data after multi-stage screening (with a results-section count of 153/345 = 44.35% high-quality on Prolific) [O’Connell et al., 2025]. Ahler et al. report that “between 25 and 35 percent” of MTurk data were of “dubious quality” across three surveys, using IP-based and behavioral indicators, emphasizing that this construct is not equivalent to other “invalid” definitions [Ahler et al., 2025].

For AI/LLM contamination, estimates depend strongly on the detection method and task. Veselovsky et al. estimate 33–46% LLM use by MTurk workers on a text-production task using keystroke detection and synthetic text classification [Veselovsky et al., 2023]. Zhang et al. report post-2022 rates of AI-generated open-ended survey responses on MTurk and Prolific that range from low single digits under a high-threshold signature-based approach (e.g., 1.37%–8.64% on MTurk; 0.51% on Prolific) to substantially higher rates under a zero-shot LLM-detector approach (e.g., 24.76%–48.71% on MTurk; 18.18% on Prolific), illustrating method sensitivity [Zhang et al., 2025].

Methodologically, these prevalence estimates should not be treated as directly interchangeable: they differ in study context (task types and incentives), operational definitions (fraud vs low-quality vs invalid), screening pipelines and thresholds, and (for AI/LLM contamination) detector choice and calibration. Accordingly, the table reports descriptive summary statistics over *point estimates* standardized as percentages of problematic responses (for broad invalid data, we report “not high-quality” = $100\% - \% \text{ high-quality}$ when necessary); range-only estimates are reported in text but excluded from the table statistics.

Table 2: Summary of empirical prevalence estimates (percent problematic) by platform and category.

Platform	Category	Studies	Estimates	Min	Max	Median	Mean	Wtd. mean
MTurk	Strict fraud/bot	2	3	35.10	64.71	56.50	52.10	49.64
Prolific	Strict fraud/bot	0	0	NA	NA	NA	NA	NA
MTurk	Broad invalid data	3	3	73.60	97.40	90.11	87.04	86.93
Prolific	Broad invalid data	2	2	32.06	55.65	43.86	43.86	41.74
MTurk	AI/LLM contamination	1	4	1.37	48.71	16.70	20.87	22.47
Prolific	AI/LLM contamination	1	2	0.51	18.18	9.35	9.35	9.35

Notes. “Studies” counts distinct PDFs contributing point estimates in each cell. “Estimates” counts distinct numeric point estimates used in the summary statistics. Broad invalid data is standardized as percent failing a study’s high-quality/validity criteria when reported as a complement. Weighted means use the study-reported sample sizes for the corresponding point estimates; some estimates (e.g., range-only reports) lack sufficient information for weighting and are excluded. Range-only estimates reported in text but excluded from the table include: 65–84% fraudulent on MTurk [Maline and Polonijo, 2025], 25–35% “dubious quality” on MTurk [Ahler et al., 2025], and 33–46% LLM use on MTurk [Veselovsky et al., 2023].

Table 3: Provisional AI-derived summary of empirical prevalence estimates (percent problematic) by platform and category; figures are based on AI-generated research digests and should be interpreted with caution.

Platform	Category	Studies	Estimates	Min	Max	Median	Mean	Wtd. mean
MTurk	Strict fraud/bot	3	4	16.70	55.00	26.75	31.30	30.84
Prolific	Strict fraud/bot	0	0	NA	NA	NA	NA	NA
MTurk	Broad invalid data	3	3	73.60	97.40	81.92	84.31	91.19
Prolific	Broad invalid data	2	2	32.10	33.62	32.86	32.86	32.86
MTurk	AI/LLM contamination	1	2	73.20	80.30	76.75	76.75	NA
Prolific	AI/LLM contamination	4	7	0.99	35.40	33.30	23.67	NA

Critical note. This table is based on **AI-generated secondary research summaries** supplied by the user, not a full manual audit of the cited articles, preprints, and platform reports. The figures should therefore be interpreted **with caution** and treated as **provisional**. They are useful for orientation and comparison, but they are **not equivalent to a verified systematic review**.

Coding rules. “Strict fraud/bot” includes percentages explicitly framed as fraudulent, suspicious, puppet, or potential bot accounts/responses. “Broad invalid data” includes percentages harmonized from low high-quality / validity retention rates (i.e., converted to problematic shares). “AI/LLM contamination” includes platform-reported AI flags, self-reported AI assistance, or inferred classifier-based LLM-use prevalence.

Included point estimates. MTurk strict fraud/bot includes 16.7% (Kennedy et al.), 18.9% (Rodriguez & Oppenheimer), 34.6% and 55.0% (Wang/Jui/Thorpe). MTurk broad invalid data includes 97.4% invalid (Webb & Tangney), 73.6% invalid derived from 26.4% high-quality (Douglas et al./JHU roundup), and 81.92% invalid derived from 18.08% retained after final screening (O’Connell et al.). Prolific broad invalid data includes 32.10% invalid derived from 67.9% high-quality (Douglas et al./JHU roundup) and 33.62% invalid derived from 66.38% retained after final screening (O’Connell et al.). MTurk AI/LLM contamination includes 80.3% and 73.2% self-reported AI-chatbot use for task completion (Christoforou et al.). Prolific AI/LLM contamination includes 0.99% platform-flagged AI-generated responses (Prolific internal audit), 13.1% and 13.4% self-reported AI-chatbot use for task completion (Christoforou et al.), 34.3% self-reported AI use on open-ended survey questions (Westwood), and 33.3%, 35.2%, and 35.4% inferred LLM-prevalence estimates in summarization tasks (Veselovsky et al.).

Weighted means. Weighted means use study-reported sample sizes where sufficiently clear from the AI summaries. Cells are marked NA where the summaries did not provide enough precise sample information, where estimates were platform-level operational figures, or where mixing fundamentally different estimands would make weighting misleading.

Excluded from summary statistics. To preserve comparability, most *range-only* figures were excluded from the summary calculations, including 33–46% MTurk LLM use (Veselovsky et al.), 15–20% MTurk VPS / non-US IPs (past audit), and similar range-only estimates. Also excluded were non-platform-specific or mixed-platform figures such as 30.55% average AI-generated responses across mixed post-2022 studies, unless platform-specific decomposition was available.

Interpretation. These estimates are highly heterogeneous because they combine different task types (surveys vs. text production), constructs (fraud, invalidity, AI assistance), and identification strategies (IP/VPS flags, post-hoc screening, behavioral traces, self-report, classifier inference, and internal platform audits). Cross-cell comparisons should therefore be treated as descriptive rather than strictly causal or directly commensurate.

2.2 mixed studies

There is no single consensus “bot/AI pollution percentage” across the corpus. Reported contamination varies widely by study context, recruitment channel, incentives, and (critically) the operational definition used (e.g., *bots* vs *fraud/suspicious* vs *LLM-assisted responses* vs *researcher-deployed agents*). Importantly some paper report rates of *suspicion* (e.g., IP-indicator flags) rather than confirmed bots/agents, whereby others report *detection performance* (sensitivity / false-positive rates) rather than prevalence

Table 4 compiles a set of explicit percentage statements found in the loaded papers, along with short context notes.

Table 4: Reported rates of bots/AI agents/LLM-mediated responding (selected examples from the loaded corpus).

Paper	Platform / sample (platform; N ; country)	Detection / classification approach	Percent(s)	Notes / caveats
[Pozzar et al., 2020]	Facebook and Twitter recruitment; $N = 271$ completed surveys within 7 hours; United States (eligible participants: English-speaking US adults diagnosed with ovarian cancer within the last 12 months)	Manual post hoc data-quality review prior to remuneration; indicators categorized as <i>suspicious</i> (potential respondent error) vs <i>fraudulent</i> (suggesting misrepresentation or automation); cases classified as fraudulent if ≥ 1 fraudulent indicator or ≥ 3 suspicious indicators; indicators included inconsistent responses to verifiable items, duplicate/open-ended responses, inattentive responding, and hidden-item triggers indicating possible automation	94.5% fraudulent; 5.5% suspicious; 16.2% showed evidence of bot automation	Bot automation was measured as one indicator of fraud (response to hidden survey items) rather than a standalone bot prevalence estimate; fraud classification rules are study-specific.
[Georgeac, 2026]	Prolific Academic; retrospective analysis of five survey samples (Samples 1–5), $N = 4,225$; United States (restricted via Prolific country location screener)	IP address analysis using IP2Location; comparison of IP-based geolocation with Qualtrics geolocation; five indicators of potential fraud (geolocation discrepancies, non-US IP location, proxy/VPN/VPS use, datacenter/hosting classification, suspicious domain characteristics)	anomalous Qualtrics geolocation prevalence increased from 4% (Sample 1) to 28% (Sample 4); depending on exclusion rule, 13–41% of responses flagged/excluded as potentially fraudulent	Indicators reflect potential fraud based on IP metadata rather than confirmed bots/LLM use; retrospective IP analysis conducted after data collection.
[Bonett et al., 2024]	Web-based COVID-19 survey (PhillyCEAL Common Survey); $N = 7,950$ completed responses; residents aged ≥ 13 years; Philadelphia, PA, United States	Two fraud-detection systems compared: (1) multilayer strategy combining automated validation, real-time exclusion rules, and post hoc verification with staff review vs (2) Qualtrics proprietary fraud detection tools (reCAPTCHA v3, RelevantID FraudScore, RelevantID DuplicateScore)	40.60% valid (multilayer); 55.21% valid (Qualtrics)	Percentages refer to responses classified as valid; remaining cases classified as fraud/invalid by each system (59.40% vs 44.79%). Fraud includes various indicators (eg, inconsistent responses, VPN/IP anomalies, duplicates) and is not limited to automated bots.
[Rodriguez and Oppenheimer, 2024]	MTurk workers recruited via CloudResearch and completing a Qualtrics survey; $N = 906$; worker country location not verified (MTurk sample)	Hand-coded free-response “identification questions” used to classify responses as bot-like vs human-like; three trained RAs coded responses. Results compared with Google reCAPTCHA v3 and with newly proposed automated bot-screening questions grounded in psychological tasks (e.g., image processing, theory-of-mind, learning, and association tasks).	18.9% potential bots (hand-coded classification); 1.7% flagged by reCAPTCHA v3	“Potential bots” are defined via authors’ coding of free-response answers and their quality-score scheme; worker country location was not verified and bot-screening protections typically available via CloudResearch were intentionally disabled to allow bot presence.
[Griffin et al., 2022]	Internet-based COVID-19 survey on LGBTQ+ populations; Wave 1 initial sample $N = 1,251$; United States	Five-step data integrity protocol: remove $< 60\%$ completion; remove low Google reCAPTCHA v3 score (reported threshold < 0.5); remove outlier completion times (reported < 5 or > 30 minutes); remove exact duplicate qualitative responses; remove conflicting/inconsistent data (e.g., demographic/ZIP inconsistencies)	61.8% removed (wave 1); 31.9% removed (wave 2); later wave: 27.4% possible bots and 72.5% valid participants	“Possible bots” are operationalized via the protocol; not necessarily confirmed automation.
[Loebenberg et al., 2023]	Remotely conducted RCT recruitment (Drink Less app); $n = 5,602$; country not specified in the extracted method snippet (UK institutions)	Address authentication and telephone verification used to detect two deception types: automated “bots” (clustered responses) vs manual deception (false information)	75.6% identified as bots (first 2 months of recruitment); 5% identified as false participants (excluding bots)	Distinguishes automated vs manual deception; location eligibility is not stated in the excerpted text.
[Shaw et al., 2025]	Experience-sampling study baseline survey; 24,053 baseline responses; country not specified in extracted snippet	Applies multiple bot/fraud detection tactics (BFDTs) with an evolving protocol; reports the fraction eliminated as likely bots or fraudulent responders	99.75% eliminated (likely bots or fraudulent)	Percent refers to elimination under BFDT criteria in a high-infiltration context.
[Claassen et al., 2026]	Facebook recruitment; $N = 1,512$; country not specified in extracted snippet; plus two LLM-driven bots (400 responses each)	Detects LLM-generated text in open narratives using a fine-tuned BERT model; bots labeled as LLM-generated (“yes”); Facebook responses given proxy label (“unclear”); prevalence is inferred from model predictions under proxy-label uncertainty	1%–3% suggested bot prevalence (by question)	Facebook responses are not ground-truth labeled (proxy “unclear”), so prevalence is inferential; the paper also reports model accuracy (97%–100%) for identifying bot responses in its setup (detection performance, not prevalence).
[Comachio et al., 2025]	Scoping review; 23 studies; multi-country (study locations vary)	Review synthesis of reported detection/countermeasures in online health recruitment; extracts whether studies encountered fraud and what proportions were excluded for fraudulent/suspicious responses	78% of studies reported encountering fraud; excluded proportions ranged 3%–94%	Review-level summary; “fraud” and exclusion thresholds vary across included studies.
[Goodrich et al., 2023]	Two incentive-based online surveys (beekeeping; Virginia farms); United States	Case-study filtering/validation (e.g., patterns in contact info, timing, and other integrity checks described by authors) used to assess fraudulent responding post hoc	96% estimated fraudulent (beekeeping survey); 72% assessed fraudulent (Virginia survey)	“Estimated/assessed” fraud depends on study-specific criteria and validation approach.
[Lin, 2026]	Conceptual/synthesis paper; reports self-reported AI use on Prolific and MTurk from cited work	Uses summarized self-report evidence (from cited studies in the paper) about participants using AI for open-ended survey questions	~33% (“one-third”) on Prolific; up to 73% on MTurk (some task types)	These percentages are self-reports attributed to external studies cited within the paper, not direct measurements from a single dataset in this manuscript.
[M. Affonso, 2025]	Multiple platforms; 1,007 assumed human participants + 526 researcher-deployed AI agents; country not specified	“Cognitive traps” derived from vision-language model constraints; reports detection performance against deployed agents plus false positives in assumed humans; also cites external prevalence claims (LLM-mediated responding)	97.1% agent detection; 4.1% false positives; also cites up to 45% LLM-mediated submissions and up to 24% continued LLM use despite instructions	The 97.1% and 4.1% are detection-performance metrics from a ground-truth agent test; the 45% and 24% are cited from other work in-text (not established as a corpus-wide prevalence estimate in this paper).

3 Measures by Platforms

3.1 Prolific

Table 5: Official Prolific documentation and announcements on handling AI-assisted and automated participation in studies.

Official source	Date / status	Threat model	What Prolific officially states	Operational implication for researchers	Source URL
Researcher Help Center: “What are authenticity checks?”	Updated 2026 (researcher documentation)	LLM-assisted responses; AI agents; fully automated bots	Prolific distinguishes two separate authenticity-check modules: <i>LLM authenticity checks</i> for free-text responses and <i>bot authenticity checks</i> for non-human or scripted behavior across question types. The documentation emphasizes that these are distinct risk classes and should be interpreted independently.	Researchers should not treat “AI use” and “bot activity” as the same construct. Study setup, interpretation, and exclusion logic should distinguish free-text mediation by LLMs from broader automated participation.	researcher-help.prolific.com/.../445144
Researcher Help Center: “What are authenticity checks?”	Updated 2026 (researcher documentation)	LLM-mediated free-text responding	Prolific states that LLM checks rely on <i>behavioral</i> indicators such as copy-paste and tab-switching, and explicitly notes that the system does <i>not</i> analyze the semantic content of the written response itself.	This positions Prolific’s approach as behavior-based authenticity assessment rather than content-based AI-text detection. For methods sections, this supports describing the intervention as telemetry/behavioral screening rather than linguistic classification.	researcher-help.prolific.com/.../445144
Researcher Help Center: “What are authenticity checks?”	Updated 2026 (researcher documentation)	False-positive risk under externally assisted tasks	Prolific advises researchers <i>not</i> to use LLM authenticity checks when a study requires participants to consult external resources, summarise outside material, or otherwise use third-party information in free-text responses.	The official guidance implies a clear boundary condition: authenticity checks are appropriate for tasks requiring original reflection or lived experience, but may be invalid for research, summarisation, or lookup-based study designs.	researcher-help.prolific.com/.../445144
Participant Help Center: “What are authenticity checks?”	Updated 2025 (participant-facing policy)	AI-generated or externally copied free-text responses	Prolific tells participants that authenticity checks are used to identify responses generated with AI tools such as ChatGPT or copied from external sources, and that flagged submissions may be reviewed by researchers.	This confirms that the policy is not merely researcher-facing infrastructure; it is also part of participant-facing conduct expectations and thus functions as a formal platform norm regarding non-original response generation.	participant-help.prolific.com/.../445031
Participant Help Center: “Why was my submission rejected?”	Updated 2025 (participant-facing policy)	Enforcement / rejection basis	Prolific states that if a researcher determines that a participant used AI tools or similar methods to complete answers, the submission may be rejected.	This is important for reporting platform governance: Prolific’s anti-AI stance is not only advisory but linked to downstream enforcement at the submission level through researcher review and rejection.	participant-help.prolific.com/.../445040
Prolific Resources: “Introducing new authenticity checks”	4 Feb 2026 (product announcement)	LLM use in text responses	Prolific reports that its LLM checks detect AI-assisted free-text responding using “15 different behaviors” and claims 98.7% precision . The announcement again frames detection as behavioral rather than semantic. In the same announcement, Prolific states that its bot checks identify AI agents or bots across the whole study and claims 100% accuracy in testing .	This source is useful when describing Prolific’s claimed technical capability, but the figure should be reported as a <i>platform-reported performance claim</i> , not as an independently audited field prevalence estimate, not a verified real-world prevalence or error-rate estimate across all studies.	...checks-beta-ensure-genuine-human-responses...
Researcher Help Center: “limitations of reCAPTCHA”	Updated 2026 (methodological guidance)	Overreliance on CAPTCHA / single-score bot screening	Prolific warns that reCAPTCHA-style bot scores should not be used as a standalone exclusion criterion and recommends interpreting such signals only as one component within a broader quality assessment framework.	This is methodologically important: Prolific’s official position favors layered, multi-factor authenticity assessment over single-indicator automated screening, especially where accessibility or geography may distort bot scores.	researcher-help.prolific.com/.../445222

Notes. The table summarizes *official Prolific sources* only (researcher help, participant help, and Prolific product/resources pages). It does not report independent prevalence estimates of bots or AI contamination on Prolific; rather, it documents the platform’s stated threat model, enforcement position, and detection architecture. Performance figures such as 98.7% precision and 100% accuracy in testing are vendor-reported claims from Prolific’s own materials and should therefore be interpreted as platform-reported evaluation outcomes rather than externally validated benchmark results. This distinction is consistent with the uploaded PDF’s conclusion that Prolific-specific strict fraud/bot prevalence evidence in the reviewed corpus is limited, while official platform documentation on AI/automation controls is more developed.

Prolific’s official position can be characterized as a *layered behavioral-authenticity approach*: it distinguishes LLM-mediated free-text assistance from full automation, relies primarily on behavioral indicators rather than semantic text inspection, permits researcher-side enforcement through flagged submissions, and explicitly discourages reliance on single-indicator tools such as reCAPTCHA alone.

3.1.1 Implications for researchers

Current official Prolific documentation indicates that the platform addresses AI-assisted and automated participation through a layered authenticity framework rather than through a single default safeguard. Specifically, Prolific distinguishes between *LLM checks*, which target AI-mediated free-text responding, and *bot checks*, which target broader non-human or scripted participation across the study. Importantly, LLM checks are **not enabled by default**; rather, they must be actively selected by the researcher during study setup on supported study infrastructures. According to Prolific’s current documentation, LLM checks are available only in contexts where the study includes *free-text or open-ended written responses* and is run on supported infrastructures, currently including *Qualtrics* and *AI Task Builder*; a Prolific product update additionally indicates support for *Gorilla’s Questionnaire Builder*. Prolific further states that these checks are relevant only when the research task requires participants to provide their *own original thoughts, opinions, or experiences*, and should *not* be used in studies without free-text questions or in designs that legitimately require participants to consult external sources, summarize external materials, or use tools outside the study environment. Accordingly, the use of these checks should be treated as a study-level design decision rather than as a universal platform-wide protection mechanism. Whereby Prolific explicitly discourages the use of single-indicator screening tools such as reCAPTCHA as a standalone exclusion criterion. For researchers, this implies two practical considerations. First, studies should not assume that all Prolific data have been screened with LLM detection unless such checks were explicitly activated. Second, authenticity checks appear most defensible in designs that require original, unaided free-text responses (@Julius Anmerkung: das macht kein Sinn, da Fragebogen/ Reaktionszeitstudien usw. problemlos KI Agenten beantworten können). Overall, Prolific’s current position is best characterized as an *opt-in, layered, behavior-based mitigation strategy that supports researcher-side enforcement, but does not eliminate the need for study-specific quality-control decisions*.

4 Bot defence Overview

This section synthesizes concrete bot detection and mitigation methods reported across the provided PDF corpus (`./context/`). Consistent with prior survey-integrity guidance, the highest-performing approach in practice is a layered, multi-signal system (rather than any single “silver bullet” check), with escalation to higher-friction challenges and (when necessary) human review [Xu et al., 2022, Bonett et al., 2024, Rilla et al., 2025]. All BibTeX entries cited in this section are included in `output/references.bib`.

4.1 Ranked methods by practical usefulness

We rank methods by (i) deployability in real systems, (ii) marginal value against adaptive adversaries, and (iii) user-friction/false-positive risk.

1. **Multi-signal risk scoring + manual review backstop** (combine weak signals; do not exclude on a single indicator) [Xu et al., 2022, Bonett et al., 2024, Maline and Polonijo, 2025, Rilla et al., 2025].
2. **Network reputation and geo-consistency checks** (VPN/proxy/VPS/datacenter detection; GeoIP mismatch; repeated coordinates) [Webster, 2023, Maline and Polonijo, 2025, Bonett et al., 2024, Pozzar et al., 2020].
3. **Unique links/tokens + recruitment URL/referrer validation** (limit link leakage and detect invalid entry paths) [Bonett et al., 2024, Comachio et al., 2025, Griffin et al., 2022].
4. **Nonindexing / reduce discoverability** (prevent search-engine discovery) [Xu et al., 2022, Karumathil and Tripathi, 2025, Comachio et al., 2025, Shaw et al., 2025].
5. **Behavioral paradata instrumentation** (mouse/keystroke timing; paste/tab-switch/window-focus; clickstream patterns) [Cuskley and Sulik, 2024, Christoforou et al., 2024, Wang et al., 2025, Rilla et al., 2025].
6. **Text similarity and templating detection** (exact/near-duplicate open-ended text; clustering) [Maline and Polonijo, 2025, Bonett et al., 2024, Rilla et al., 2025].
7. **Screener/main and cross-item consistency checks** (repeat verifiable items; logical constraints; persona consistency) [Bonett et al., 2024, Pozzar et al., 2020, Maline and Polonijo, 2025, Westwood, 2025].
8. **Device/browser linkage signals** (fingerprinting/cookies/local storage to detect repeat/linked submissions) [Webster, 2023, Wang et al., 2025, Jaffe et al., 2026, Shaw et al., 2025].
9. **Rate limiting and submission caps** (throttle bursts; per-IP/device quotas; cooldowns) [Shaw et al., 2025, Rilla et al., 2025].
10. **Step-up challenges** (CAPTCHA, honeypots, and real-effort micro-challenges triggered by risk) [Bonett et al., 2024, Höhne et al., 2025, Karumathil and Tripathi, 2025, Rilla et al., 2025, Rodriguez and Oppenheimer, 2024].

4.2 Method catalog (merged across papers) by category

Behavioral / paradata

Completion-time and per-page timing thresholds (“speeders”) [Detection]. Flag implausibly fast completions or page dwell times. Strength: low cost and widely available; Weakness: can be evaded by slowing down and may exclude legitimately fast humans [Xu et al., 2022, Webb and Tangney, 2024, Comachio et al., 2025, Höhne et al., 2025, Agley, 2025].

Interaction-trace profiling (mouse/keystrokes/paste/tab-switch/focus) [Detection; sometimes Mitigation]. Record fine-grained interaction patterns and detect automation-like traces (e.g., copy/paste, unnatural timing regularities). Strength: harder to spoof at scale than single-item checks; Weakness: sophisticated agents can simulate interaction patterns [Cuskley and Sulik, 2024, Christoforou et al., 2024, Wang et al., 2025, Rilla et al., 2025].

Low-effort response patterns (straightlining/low variance/long strings) [Detection]. Strength: cheap post-hoc screen; Weakness: not bot-specific [Douglas et al., 2023, Lebrun et al., 2024a].

Speed-bump and effort tasks [Mitigation; sometimes Detection]. Insert items intended to slow automation or require real effort. Strength: raises attacker cost; Weakness: participant burden and accessibility concerns [Comachio et al., 2025, Karumathil and Tripathi, 2025, Rodriguez and Oppenheimer, 2024].

Attention checks (instructed response items; bogus/absurd items) [Detection]. Standard attention-check families used as one signal among many. Strength: simple and widely used; Weakness: can misclassify misunderstanding and can be solved by more capable automators [Muszyński, 2023, Karumathil and Tripathi, 2025, Westwood, 2025].

Logic puzzles, instruction-following tasks, and “impossible” sanity checks [Detection]. Gates or flags based on puzzles/instructions and implausible claims. Strength: high face validity; Weakness: can be passed by synthetic agents and should not be used alone [Westwood, 2025].

Reverse shibboleth tasks [Detection]. Detect “too-competent” AI-like performance on tasks that most humans would not complete. Strength: targets agentic AI; Weakness: can be evaded via strategic refusal patterns [Westwood, 2025, Lin, 2026].

Consent-comprehension quiz gate [Both]. Short quiz on consent details; terminate after repeated failures. Strength: early, low-cost filter; Weakness: added friction [Webb and Tangney, 2024].

Perceptual/cognitive trap tasks for vision-enabled agents [Detection]. Use specialized perceptual traps and thresholded failure counts. Strength: reported higher sensitivity than conventional checks in some evaluations; Weakness: false positives depend on design [M. Affonso, 2025].

Network / traffic

VPN/proxy/VPS/datacenter detection and blocking [Both]. Score or block submissions from suspicious networks, especially in geo-restricted studies. Strength: high value against fraud farms; Weakness: imperfect geolocation and well-instrumented adversaries can route through residential networks [Webster, 2023, Maline and Polonijo, 2025, Bonett et al., 2024, Pozzar et al., 2020, Georgeac, 2026].

Duplicate IP and burst pattern checks [Detection; sometimes Mitigation]. Flag excessive submissions per IP and short inter-arrival times; cap submissions. Strength: simple; Weakness: rotating IPs and shared networks [Xu et al., 2022, Webster, 2023, Bonett et al., 2024].

GeoIP consistency and repeated coordinate patterns [Detection]. Detect mismatches between claimed eligibility and inferred location, or repeated coordinate clusters. Strength: useful against centralized farms; Weakness: location inference is noisy [Bonett et al., 2024, Maline and Polonijo, 2025].

Rate limiting / throttling / cooldowns [Mitigation]. Reduce attack throughput. Strength: immediate impact; Weakness: can affect legitimate bursts [Shaw et al., 2025, Rilla et al., 2025].

Device/browser fingerprinting and repeat-submission controls

Cookie-based “prevent multiple submissions” [Mitigation; also Detection]. Common platform setting that blocks repeats via cookies. Strength: very easy; Weakness: evasion via clearing cookies/incognito [Xu et al., 2022, Webster, 2023, Karumathil and Tripathi, 2025, Maline and Polonijo, 2025].

Device/browser linkage (fingerprints, local storage identifiers) [Both]. Use stable (privacy-aware) linkage signals to identify repeat/linked submissions across IP rotation. Strength: better than IP alone; Weakness: anti-fingerprinting measures and sophisticated mimicry [Webster, 2023, Wang et al., 2025, Jaffe et al., 2026, Shaw et al., 2025].

CAPTCHA / challenge-response / trap fields

CAPTCHA (incl. risk-scored variants) [Both]. Widely deployed baseline filter. Strength: easy integration; Weakness: several papers report limited effectiveness against more capable bots and potential user friction [Bonett et al., 2024, Höhne et al., 2025, Rilla et al., 2025].

Honeypot/invisible trap fields [Both]. Hidden fields/items that humans do not fill. Strength: near-zero friction; Weakness: can be bypassed and may not challenge stronger bots [Goodrich et al., 2023, Bonett et al., 2024, Höhne et al., 2025, Rilla et al., 2025].

Real-effort micro-challenges (logic/image/free-response) [Mitigation; sometimes Detection]. “Bot-screening” items grounded in psychological or cognitive tasks. Strength: can outperform standard CAPTCHAs in some settings; Weakness: labor/burden and an arms race with more capable models [Rodriguez and Oppenheimer, 2024, Karumathil and Tripathi, 2025, Westwood, 2025].

Content / linguistic signals

Open-ended coherence/meaningfulness screening [Detection]. Require explanation items and assess for nonsensical or copied content. Strength: strong against naive bots; Weakness: advanced generators can produce plausible text [Xu et al., 2022, Pozzar et al., 2020, Maline and Polonijo, 2025, Shaw et al., 2025].

Duplicate/templated text detection and clustering [Detection]. Detect repeated strings and near-duplicates across submissions. Strength: highly effective against template reuse; Weakness: paraphrasing reduces sensitivity [Maline and Polonijo, 2025, Bonett et al., 2024, Rilla et al., 2025].

AI-generated text detectors / stylometry features [Detection]. Apply automated detectors as one signal among many. Strength: scalable triage; Weakness: false positives and rapid adversarial adaptation [Lebrun et al., 2024a, Claassen et al., 2026, Zhang et al., 2025, Rilla et al., 2025].

Anomaly detection and ensembles

Rule-based “impossible/inconsistent” checks [Detection]. Cross-validate screener vs main survey, repeat verifiable items, and enforce logical constraints. Strength: high precision when verifiable; Weakness: requires careful design [Bonett et al., 2024, Pozzar et al., 2020, Maline and Polonijo, 2025].

Multi-signal clustering / anomaly detection [Detection]. Fuse behavioral, network, device, and text features into anomaly models or ensembles. Strength: robust to adaptive evasion; Weakness: requires tuning/evaluation and is less interpretable [Wang et al., 2025, Pinzón et al., 2024, Rilla et al., 2025].

Consecutive submissions and burst-timing indicators [Detection]. Flag near-simultaneous starts/finishes and very short inter-arrival times as one indicator of coordinated fraud/bot activity. Strength: strong signal in field reports; Weakness: legitimate simultaneous starts are possible and require combination logic [Pinzón et al., 2024, Shaw et al., 2025].

Time-of-day anomaly indicators [Detection]. Use unusual submission windows as one risk feature. Strength: can separate farm traffic from typical participation; Weakness: time zones and shift workers complicate thresholds [Pinzón et al., 2024].

Cross-provider IP/Geo validation [Detection]. Compare geolocation/proxy fields across providers and flag large discrepancies. Strength: operationalizes provider disagreement; Weakness: geolocation uncertainty remains fundamental [Georgeac, 2026].

Identity / verification / access control

Unique links/tokens and recruitment-source validation [Mitigation; also Detection]. Strength: very deployable; Weakness: link leakage remains possible [Bonett et al., 2024, Comachio et al., 2025, Griffin et al., 2022].

Platform identity signals (worker IDs, reputation thresholds, exclusion lists) [Both]. Use platform-level IDs and prior-quality filters where available. Strength: low friction; Weakness: multiple accounts and shifting platform conditions [Webster, 2023, Maline and Polonijo, 2025, Douglas et al., 2023].

Email/phone verification (step-up) [Mitigation]. Strength: raises attacker cost; Weakness: adds friction and equity/access concerns [Xu et al., 2022, Comachio et al., 2025].

Email-handle heuristic scoring [Detection]. Score email handles (e.g., length/digits/caps patterns) and flag suspicious addresses. Strength: reported as a high-performing indicator in some settings; Weakness: adaptive adversaries can generate more human-like patterns [Pinzón et al., 2024, Goodrich et al., 2023].

Third-party IP+email risk scoring (e.g., MinFraud RiskScore) [Detection; optional step-up]. Use risk scores as one feature (or to trigger additional verification). Strength: high predictive value at some thresholds in reported evaluations; Weakness: performance varies by context [Pinzón et al., 2024].

Human moderation and operational tactics

Manual review and adjudication using clear rubrics [Both]. Strength: best backstop for edge cases; Weakness: labor-intensive and subjective [Goodrich et al., 2023, Bonett et al., 2024, Maline and Polonijo, 2025].

Post-submission verification for payment/identity (high-friction step-up) [Both]. Use follow-up contact/verification tied to incentives; refusal/no-response can be a strong risk signal. Strength: deterrent; Weakness: privacy/accessibility trade-offs [Shaw et al., 2025].

Dynamic protocols and rotating checks [Mitigation]. Change placement/type of checks during fielding to disrupt attacker adaptation. Strength: breaks static evasion; Weakness: increases operational complexity [Shaw et al., 2025, Rilla et al., 2025].

Graph/social methods (platform-scale)

Graph-structural and synchronized-behavior detection [Detection]. Use interaction graphs and structural/community signals to detect coordinated bots (primarily social platforms). Strength: hard to fake at scale; Weakness: requires graph data and domain-specific engineering [Peng et al., 2024].

4.3 Top-10 lists and a recommended defense stack

Top 10 most effective detection methods. Multi-signal scoring; VPN/proxy + GeoIP checks; text similarity clustering; paradata instrumentation; timing outliers; cross-item consistency checks; device/browser linkage; open-ended coherence screening; ensemble anomaly detection; graph-structural detection (when applicable) [Xu et al., 2022, Bonett et al., 2024, Maline and Polonijo, 2025, Webster, 2023, Wang et al., 2025, Shaw et al., 2025, Peng et al., 2024].

Top 10 most effective mitigation methods. Nonindexing; unique links/tokens; prevent multiple submissions baseline; rate limiting; network reputation blocking/step-up; risk-triggered CAPTCHA/honeypots; real-effort micro-challenges; exclusion lists; dynamic protocols; selective verification for high-risk flows [Xu et al., 2022, Karumathil and Tripathi, 2025, Comachio et al., 2025, Bonett et al., 2024, Rilla et al., 2025, Shaw et al., 2025].

Best combined defense stack (low friction to high friction). (1) Always-on: nonindexing; unique link tokens; prevent multiple submissions; basic rate limiting. (2) Risk scoring: VPN/proxy/datacenter + GeoIP checks; device/browser linkage; paradata logging; text similarity clustering; burst/consecutive-submission indicators. (3) Step-up: CAPTCHA/honeypots and real-effort challenges only for high-risk sessions; selective verification for incentive payout in high-risk flows. (4) Backstop: cross-item consistency repeats + targeted manual review, with dynamic/rotated protocols to reduce attacker learning [Xu et al., 2022, Bonett et al., 2024, Webster, 2023, Maline and Polonijo, 2025, Pinzón et al., 2024, Rilla et al., 2025, Shaw et al., 2025].

5 Catalog of bot methods (merged across papers)

This section synthesizes concrete, corpus-reported methods used to detect survey bots and to deter, disrupt, or defend against automated and fraudulent responders. Methods are merged across papers when they are functionally equivalent. We label each method as *detection*, *mitigation*, or *both*. Where the literature reports evasion (or plausibly suggests it), we note it explicitly.

5.0.1 Behavioral

Completion-time thresholds / speeding flags **Type:** detection. **Description:** flag unusually fast (or otherwise implausible) survey/page/item completion times. **Evidence:** Storozuk et al. [2020], Griffin et al. [2022], Shaw et al. [2025], Pinzón et al. [2024], Höhne et al. [2025], Karumathil and Tripathi [2025]. **Strengths:** low cost; available in most platforms; useful for early triage. **Weaknesses/evasion:** sophisticated bots/agents can slow down; may false-positive fast readers or mobile users.

Burst/velocity patterns (consecutive submissions; overlapping timestamps) **Type:** detection. **Description:** detect near-simultaneous starts/finishes or high-frequency submission bursts. **Evidence:** Xu et al. [2022], Shaw et al. [2025], Pinzón et al. [2024], Karumathil and Tripathi [2025]. **Strengths:** strong signal for attack waves in case studies. **Weaknesses/evasion:** bursty legitimate recruitment can resemble attacks; attackers can distribute submissions over time.

Response-pattern heuristics (straightlining/long-string; reverse-item inconsistency) **Type:** detection. **Description:** flag invariant responding or inconsistent patterns on items designed to detect low-effort responding. **Evidence:** Storozuk et al. [2020], Shaw et al. [2025], Pinzón et al. [2024], Karumathil and Tripathi [2025]. **Strengths:** cheap; catches some low-effort automation/carelessness. **Weaknesses/evasion:** sophisticated bots can randomize plausibly; some humans are legitimately patterned.

Logic/consistency checks (internal contradictions; repeated items) **Type:** detection. **Description:** embed cross-item constraints and repeated questions to surface contradictions. **Evidence:** Pozzar et al. [2020], Xu et al. [2022], Griffin et al. [2022], Shaw et al. [2025], Pinzón et al. [2024], Karumathil and Tripathi [2025]. **Strengths:** study-specific; raises attacker design cost. **Weaknesses/evasion:** careful fraudsters can answer consistently; requires design effort and pretesting.

Item-nonresponse / page-interaction anomalies **Type:** detection. **Description:** flag unusual interaction traces (e.g., only first item per page answered). **Evidence:** Höhne et al. [2025]. **Strengths:** can catch simple rule-based bots. **Weaknesses/evasion:** brittle; less effective against modern AI-based bots.

Interaction telemetry (mouse/keyboard/typing/cursor; copy-paste; window switching) **Type:** detection. **Description:** instrument richer behavior streams beyond final answers. **Evidence:** Höhne et al. [2025], Jaffe et al. [2026], Lin [2026], M. Affonso [2025], Karumathil and Tripathi [2025]. **Strengths:** can reveal automation/mediation patterns that pass answer-only checks. **Weaknesses/evasion:** privacy/consent considerations; sophisticated attackers can spoof or replay realistic telemetry [Lin, 2026].

5.0.2 Network

IP capture + geolocation and eligibility checks (incl. self-report mismatch) **Type:** detection (sometimes both when used to block). **Description:** compare IP-derived location/provider to study eligibility and to self-reported location. **Evidence:** Pozzar et al. [2020], Storozuk et al. [2020], Griffin et al. [2022], Shaw et al. [2025], Bonett et al. [2024], Pinzón et al. [2024], Comachio et al. [2025]. **Strengths:** useful auxiliary signal; catches obvious ineligible traffic. **Weaknesses/evasion:** VPN/VPS; geolocation error; shared networks.

VPN/proxy/VPS detection services **Type:** both. **Description:** score/deny traffic from known proxy infrastructure. **Evidence:** Pozzar et al. [2020], Shaw et al. [2025], Pinzón et al. [2024], Karumathil and Tripathi [2025]. **Strengths:** operationally convenient. **Weaknesses/evasion:** false positives for legitimate VPN users; attackers can rotate or use residential IPs.

Recruitment URL / referrer validation **Type:** both. **Description:** ensure traffic arrives via intended channels/links. **Evidence:** Griffin et al. [2022], Bonett et al. [2024], Comachio et al. [2025]. **Strengths:** reduces exposure from link sharing. **Weaknesses/evasion:** spoofable; channel shifts can confuse interpretation.

Email/IP blacklists **Type:** detection. **Description:** block/flag entries matching known bad infrastructure. **Evidence:** Pinzón et al. [2024]. **Strengths:** easy to add. **Weaknesses/evasion:** may be ineffective or lag attacks [Pinzón et al., 2024].

5.0.3 Device / browser fingerprinting

Cookies / “prevent ballot box stuffing” **Type:** mitigation (weak detection). **Description:** prevent repeated submissions from the same browser/session. **Evidence:** Griffin et al. [2022], Comachio et al. [2025], Karumathil and Tripathi [2025]. **Strengths:** low friction; stops naive repeats. **Weaknesses/evasion:** incognito, cookie clearing, multi-device.

Browser/device “meta-information” screening **Type:** detection. **Description:** flag suspicious or inconsistent device/browser configurations. **Evidence:** Shaw et al. [2025]. **Strengths:** can catch some automated/farm setups. **Weaknesses/evasion:** modern bots can mimic realistic fingerprints.

Device fingerprinting / duplicate-device scoring (vendor tools) **Type:** both. **Description:** link repeated responders across sessions/devices using fingerprinting/duplicate scores. **Evidence:** Bonett et al. [2024], Jaffe et al. [2026]. **Strengths:** helps de-duplicate at scale. **Weaknesses/evasion:** privacy constraints; attackers can diversify devices.

5.0.4 CAPTCHA / challenge–response

CAPTCHA / reCAPTCHA (incl. score thresholds) **Type:** both. **Description:** gate entry or flag low-confidence traffic using CAPTCHA variants (including reCAPTCHA v3 scores). **Evidence:** Pozzar et al. [2020], Xu et al. [2022], Griffin et al. [2022], Loebenberg et al. [2023], Bonett et al. [2024], Rodriguez and Oppenheimer [2024], Höhne et al. [2025], Comachio et al. [2025]. **Strengths:** low-cost first gate; in some trials, adding CAPTCHA coincided with eliminating a bot wave [Loebenberg et al., 2023]. **Weaknesses/evasion:** modern bots and human-assist services can pass; several papers caution CAPTCHAs alone are insufficient [Rodriguez and Oppenheimer, 2024, Höhne et al., 2025].

Real-effort / uncommon task batteries **Type:** both. **Description:** add tasks intended to raise automation cost (e.g., bespoke cognitive tasks) or “speed bumps.” **Evidence:** Storozuk et al. [2020], Karumathil and Tripathi [2025], Comachio et al. [2025]. **Strengths:** increases friction for simple bots. **Weaknesses/evasion:** accessibility burden; effectiveness varies; some tactics (e.g., “text as image”) are reported as ineffective in practice [Storozuk et al., 2020].

5.0.5 Graph / social

Graph-structural bot detection (platform context) **Type:** detection. **Description:** detect bot communities via multi-relational graphs and structural information/entropy optimization. **Evidence:** Peng et al. [2024]. **Strengths:** interpretable, label-free detection for social networks. **Weaknesses/evasion:** requires graph access and platform instrumentation; not directly portable to standalone surveys.

5.0.6 Content / linguistic

Open-ended response screening (nonsense/duplicates/illogical) **Type:** detection. **Description:** use open-ended probes to detect irrelevant, duplicated, or nonsensical text; often combined with manual review. **Evidence:** Storozuk et al. [2020], Griffin et al. [2022], Goodrich et al. [2023], Bonett et al. [2024], Shaw et al. [2025], Pinzón et al. [2024], Comachio et al. [2025]. **Strengths:** high signal against low-effort attacks; duplicates are particularly diagnostic. **Weaknesses/evasion:** LLMs can generate plausible text; review can be costly.

LLM-generated text detection on open narratives **Type:** detection. **Description:** train classifiers on narrative responses to distinguish bot/LLM-generated text. **Evidence:** Claassen et al. [2026], Lebrun et al. [2024b]. **Strengths:** strong performance can be reported in constrained settings [Claassen et al., 2026]. **Weaknesses/evasion:** can degrade on new prompts/domains; tool reliability and false positives remain concerns [Lebrun et al., 2024b].

Prompt-based “trap” strings aimed at LLM-following **Type:** detection (experimental). **Description:** embed instructions intended to elicit telltale bot/LLM behavior. **Evidence:** Höhne et al. [2025]. **Strengths:** targets a specific threat model (LLM-following). **Weaknesses/evasion:** limited validation; could contaminate responses or be bypassed by robust agents.

5.0.7 Anomaly detection (statistical / model-based)

Multivariate outlier screens (e.g., Mahalanobis distance) **Type:** detection. **Description:** flag unusual response vectors using multivariate distances/correlation patterns. **Evidence:** Storozuk et al. [2020]. **Strengths:** scalable; detects non-obvious anomalies. **Weaknesses/evasion:** assumptions may fail under adaptive bots; can exclude unusual-but-valid respondents.

Multi-indicator scoring pipelines (“strikes”; staged filters) **Type:** both. **Description:** combine multiple weak signals into decision rules, often with dynamic updates during fielding. **Evidence:** Bonett et al. [2024], Shaw et al. [2025], Karumathil and Tripathi [2025]. **Strengths:** more robust than single checks; supports operations and auditing. **Weaknesses/evasion:** requires tuning/maintenance; attackers adapt.

Latent-class / mixture modeling for contamination **Type:** detection/analysis. **Description:** model a latent “contaminated” class to quantify uncertainty and sensitivity. **Evidence:** Karumathil and Tripathi [2025], Lin [2026]. **Strengths:** supports contamination-aware inference. **Weaknesses/evasion:** higher complexity; still depends on measurable signals.

5.0.8 Identity / verification

Email screening (patterns; disposable domains; bounce checks) **Type:** both. **Description:** detect suspicious email structures and require verifiable contact details. **Evidence:** Storozuk et al. [2020], Goodrich et al. [2023], Bonett et al. [2024], Shaw et al. [2025], Pinzón et al. [2024]. **Strengths:** practical and often high-yield in field reports. **Weaknesses/evasion:** attackers can generate realistic addresses; privacy considerations.

Phone verification / dual contact verification **Type:** both. **Description:** require and verify a phone number (and optionally cross-check via call/SMS); treat non-response as a fraud signal. **Evidence:** Loebenberg et al. [2023], Shaw et al. [2025], Xu et al. [2022], Comachio et al. [2025]. **Strengths:** high assurance; strong deterrent. **Weaknesses/evasion:** high friction; may reduce inclusion; operational cost.

Address authentication / mailed incentives **Type:** both. **Description:** verify physical address or mail incentives to raise the cost of fraud. **Evidence:** Pozzar et al. [2020], Loebenberg et al. [2023], Comachio et al. [2025]. **Strengths:** strong deterrent for incentive-motivated attacks. **Weaknesses/evasion:** logistical burden; privacy and equity trade-offs.

Video / live checks **Type:** both. **Description:** identity verification via video or synchronous contact. **Evidence:** Comachio et al. [2025]. **Strengths:** very high assurance. **Weaknesses/evasion:** highest friction; ethics/privacy constraints.

Vetted panels / platform-level vetting **Type:** mitigation. **Description:** recruit from panels with identity controls and fraud monitoring rather than open links. **Evidence:** Jaffe et al. [2026], Lin [2026], Karumathil and Tripathi [2025]. **Strengths:** shifts defense upstream. **Weaknesses/evasion:** cost; coverage/representativeness constraints.

Unique payment codes; delayed payout until screening completes **Type:** mitigation (and a detection lever when verification fails). **Description:** tie compensation to passing checks, often with manual review and non-automatic payment. **Evidence:** Griffin et al. [2022], Xu et al. [2022], Karumathil and Tripathi [2025], Comachio et al. [2025]. **Strengths:** reduces attacker ROI; supports adjudication. **Weaknesses/evasion:** slows payments; may affect participation.

5.0.9 Human moderation

Manual review / adjudication (often for open-ended + metadata) **Type:** detection. **Description:** human screening of borderline cases using multiple signals and documentation. **Evidence:** Shaw et al. [2025], Bonett et al. [2024], Pinzón et al. [2024], Comachio et al. [2025]. **Strengths:** flexible; catches novel attacks. **Weaknesses/evasion:** expensive; subjective without clear protocols.

5.0.10 Rate limiting

The loaded corpus contains relatively little explicit discussion of classic web-platform rate limiting (e.g., throttling by IP/session). Closely related tactics in this literature include cookie-based repeat prevention, individualized links, and burst/velocity detection [Griffin et al., 2022, Pinzón et al., 2024].

5.1 Practical ranking (real deployments)

Based on reported field performance and operational feasibility in the corpus, the most practically useful methods tend to be (i) access control (avoid public links; individualized links; prevent indexing), (ii) high-assurance verification for high-risk contexts, and (iii) multi-indicator pipelines that combine timing/velocity, contact/identity signals, and content checks, with manual adjudication for edge cases [Xu et al., 2022, Loebenberg et al., 2023, Shaw et al., 2025, Bonett et al., 2024, Comachio et al., 2025].

5.2 Top 10 lists and recommended defense stack

Top 10 most effective bot detection methods (deployment-useful).

1. Multi-indicator/staged filtering (“strikes”; pipelines) [Shaw et al., 2025, Bonett et al., 2024].
2. Burst/velocity detectors (consecutive submissions; timestamp overlap) [Pinzón et al., 2024, Shaw et al., 2025].
3. Failed/ignored phone verification as a fraud signal [Shaw et al., 2025, Loebenberg et al., 2023].
4. Email pattern screening + domain/bounce checks [Pinzón et al., 2024, Storozuk et al., 2020, Bonett et al., 2024].
5. De-duplication across IP/email/phone/free-text [Goodrich et al., 2023, Bonett et al., 2024].
6. Open-ended probe screening (duplicates/illogical text) + manual review [Griffin et al., 2022, Shaw et al., 2025].
7. reCAPTCHA scoring (as one feature, not a sole decision rule) [Bonett et al., 2024, Griffin et al., 2022].
8. Location mismatch (self-report vs IP-derived) [Pozzar et al., 2020, Shaw et al., 2025].
9. LLM-text classifiers for narrative responses (when open text is central) [Claassen et al., 2026, Lebrun et al., 2024b].
10. Behavioral telemetry (copy-paste/window switching/typing traces), where ethically and technically feasible [Jaffe et al., 2026, Lin, 2026].

Top 10 most effective bot mitigation methods (deployment-useful).

1. Avoid public survey links; use sign-up then individualized links [Xu et al., 2022, Storozuk et al., 2020, Pozzar et al., 2020].
2. Prevent indexing / search-engine blocking [Karumathil and Tripathi, 2025, Comachio et al., 2025].
3. Use vetted panels/platforms with upstream controls when possible [Jaffe et al., 2026, Lin, 2026].
4. High-assurance verification gates for high-risk studies (phone/address/video) [Loebenberg et al., 2023, Comachio et al., 2025].
5. Delay/withhold payouts until screening completes; avoid automatic payment [Griffin et al., 2022, Xu et al., 2022].
6. Incentive redesign (downplay, raffle, physical mailing) to reduce attacker ROI [Pozzar et al., 2020, Griffin et al., 2022, Comachio et al., 2025].
7. CAPTCHA/reCAPTCHA as a first gate (not alone) [Loebenberg et al., 2023, Rodriguez and Oppenheimer, 2024].
8. Cookie-based repeat prevention [Griffin et al., 2022, Comachio et al., 2025].
9. Rotate/randomize checks; dynamic protocols during fielding [Shaw et al., 2025, Karumathil and Tripathi, 2025].
10. “Speed bumps” / imposed delays to add friction and enable timing signals [Comachio et al., 2025, Karumathil and Tripathi, 2025].

Best combined defense stack for a real-world platform (corpus-backed).

1. **Reduce exposure:** disable public links; require sign-up; distribute individualized links; block indexing [Xu et al., 2022, Karumathil and Tripathi, 2025].
2. **Front-door gate:** CAPTCHA/reCAPTCHA + cookies; log IP and basic metadata [Griffin et al., 2022, Loebenberg et al., 2023, Bonett et al., 2024].
3. **In-survey instrumentation:** page-level timing; contradiction checks; at least one open-ended probe; rotate checks [Shaw et al., 2025, Pinzón et al., 2024].
4. **Back-end pipeline:** burst detectors; de-duplication; email screening; multi-indicator scoring with human adjudication for edge cases [Pinzón et al., 2024, Bonett et al., 2024, Shaw et al., 2025].
5. **High-assurance escalation:** for high-fraud waves or high-value studies, require phone/address (or, rarely, video) verification; delay payouts until verification [Comachio et al., 2025, Loebenberg et al., 2023].

6 Detect bots and defend

Similar to previous studies M. Affonso [2025], deployed AI agents to act as participants in a study or online environment so we could test methods, measure behavior, and evaluate approaches to detect and defend against bots.

References

- Margaret Husted, Anna Dowrick, Robert Porter, Maria Velo Higuera, Carly Whitmore, Jane Evered, Megan Kennedy, and Shannon D. Scott. Imposter Participants in Synchronous Qualitative Research: A Systematic Scoping Review. *International Journal of Qualitative Methods*, 24:16094069251342542, April 2025. ISSN 1609-4069. doi:10.1177/16094069251342542. URL <https://doi.org/10.1177/16094069251342542>.
- Natalia Pinzón, Vikram Koundinya, Ryan E. Galt, William O'R Dowling, Marcela Baukloh, Namah C. Taku-Forchu, Tracy Schohr, Leslie M. Roche, Samuel Ikendi, Mark Cooper, Lauren E. Parker, and Tapan B. Pathak. AI-powered fraud and the erosion of online survey integrity: an analysis of 31 fraud detection strategies. *Frontiers in Research Metrics and Analytics*, 9, December 2024. ISSN 2504-0537. doi:10.3389/frma.2024.1432774. URL <https://www.frontiersin.org/journals/research-metrics-and-analytics/articles/10.3389/frma.2024.1432774/full>.
- Yanfeng Xu, Sarah Pace, Jaeseung Kim, Aidyn Iachini, L Bailey King, Theresa Harrison, Dana DeHart, Sue E Levkoff, Teri A Browne, Ashlee A Lewis, Gina M Kunz, Melissa Reitmeier, R Karen Utter, and Melissa Simone. Threats to Online Surveys: Recognizing, Detecting, and Preventing Survey Bots. *Social Work Research*, 46(4):343–350, December 2022. ISSN 1070-5309. doi:10.1093/swr/svac023. URL <https://doi.org/10.1093/swr/svac023>.
- Damien Ridge, Laurna Bullock, Hilary Causer, Tamsin Fisher, Samantha Hider, Tom Kingstone, Lauren Gray, Ruth Riley, Nina Smyth, Victoria Silverwood, Johanna Spiers, and Jane Southam. 'Imposter participants' in online qualitative research, a new and increasing threat to data integrity? *Health Expectations : An International Journal of Public Participation in Health Care and Health Policy*, 26(3):941–944, February 2023. ISSN 1369-6513. doi:10.1111/hex.13724. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC10154888/>.
- Pakhi Sharma, Steven M. McPhail, Sanjeewa Kularatna, Sameera Senanayake, and Bridget Abell. Navigating the challenges of imposter participants in online qualitative research: lessons learned from a paediatric health services study. *BMC Health Services Research*, 24(1):724, June 2024. ISSN 1472-6963. doi:10.1186/s12913-024-11166-x. URL <https://doi.org/10.1186/s12913-024-11166-x>.
- Alan Santinele Martino, Arielle Perrotta, and Brenna Janet McGillion. Who can you trust these days?: Dealing with imposter participants during online recruitment and data collection. *Qualitative Research*, 24(5):1291–1301, October 2024. ISSN 1468-7941. doi:10.1177/14687941231224591. URL <https://doi.org/10.1177/14687941231224591>.
- Raluca Rilla, Tobias Werner, Hiromu Yakura, Iyad Rahwan, and Anne-Marie Nussberger. Recognising, Anticipating, and Mitigating LLM Pollution of Online Behavioural Research, November 2025. URL <http://arxiv.org/abs/2508.01390>. arXiv:2508.01390 [cs].
- Jon Agley. Planning for New Threats to Online Research Data Validity: The Issue of Computer-Using Agents. *Evaluation & the Health Professions*, page 01632787251367407, August 2025. ISSN 0163-2787. doi:10.1177/01632787251367407. URL <https://doi.org/10.1177/01632787251367407>.
- Sean J. Westwood. The potential existential threat of large language models to online survey research. *Proceedings of the National Academy of Sciences*, 122(47):e2518075122, November 2025. doi:10.1073/pnas.2518075122. URL <https://www.pnas.org/doi/10.1073/pnas.2518075122>.
- Rachel Pozzar, Marilyn J. Hammer, Meghan Underhill-Blazey, Alexi A. Wright, James A. Tulskey, Fangxin Hong, Daniel A. Gundersen, and Donna L. Berry. Threats of Bots and Other Bad Actors to Data Quality Following Research Participant Recruitment Through Social Media: Cross-Sectional Questionnaire. *Journal of Medical Internet Research*, 22(10):e23021, October 2020. doi:10.2196/23021. URL <https://www.jmir.org/2020/10/e23021>. Company: Journal of Medical Internet Research Distributor: Journal of Medical Internet Research Institution: Journal of Medical Internet Research Label: Journal of Medical Internet Research.
- Marissa N. Maline and Andrea N. Polonijo. Unmasking survey fraud: investigating data quality issues in an MTurk sample. *International Journal of Social Research Methodology*, 0(0):1–8, December 2025. ISSN 1364-5579. doi:10.1080/13645579.2025.2601994. URL <https://doi.org/10.1080/13645579.2025.2601994>. _eprint: <https://doi.org/10.1080/13645579.2025.2601994>.
- Stephen Bonett, Willey Lin, Patrina Sexton Topper, James Wolfe, Jesse Golinkoff, Aayushi Deshpande, Antonia Villaruel, and José Bauermeister. Assessing and Improving Data Integrity in Web-Based Surveys: Comparison of Fraud Detection Systems in a COVID-19 Study. *JMIR Formative Research*, 8(1):e47091, January 2024. doi:10.2196/47091. URL <https://formative.jmir.org/2024/1/e47091>. Company: JMIR Formative Research Distributor: JMIR Formative Research Institution: JMIR Formative Research Label: JMIR Formative Research.
- Joshua Claassen, Jan Karem Höhne, Ruben Bach, and Anna-Carolina Haensch. Identifying Bots Through LLM-Generated Text in Open Narrative Responses: A Proof-of-Concept Study. *Social Science Computer Review*,

- page 08944393251408022, January 2026. ISSN 0894-4393. doi:10.1177/08944393251408022. URL <https://doi.org/10.1177/08944393251408022>.
- Benjamin Lebrun, Sharon Temtsin, Andrew Vonasch, and Christoph Bartneck. Detecting the corruption of on-line questionnaires by artificial intelligence. *Frontiers in Robotics and AI*, 10, February 2024a. ISSN 2296-9144. doi:10.3389/frobt.2023.1277635. URL <https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2023.1277635/full>.
- Shengqian Wang, Israt Jahan Jui, and Julie Thorpe. Is Crowdsourcing a Puppet Show? Detecting a New Type of Fraud in Online Platforms. In *Proceedings of the New Security Paradigms Workshop*, NSPW '24, pages 84–95, New York, NY, USA, January 2025. Association for Computing Machinery. ISBN 979-8-4007-1128-2. doi:10.1145/3703465.3703472. URL <https://dl.acm.org/doi/10.1145/3703465.3703472>.
- Andie Storozuk, Marilyn Ashley, Véronique Delage, and Erin A. Maloney. Got Bots? Practical Recommendations to Protect Online Survey Data from Bot Attacks. *The Quantitative Methods for Psychology*, 16(5):472–481, May 2020. ISSN 2292-1354. doi:10.20982/tqmp.16.5.p472. URL <http://www.tqmp.org/RegularArticles/vol16-5/p472>.
- Jan Karem Höhne, Joshua Claassen, Saijal Shahania, and David Broneske. Bots in web survey interviews: A showcase. *International Journal of Market Research*, 67(1):3–12, January 2025. ISSN 1470-7853. doi:10.1177/14707853241297009. URL <https://doi.org/10.1177/14707853241297009>.
- Shalom N. Jaffe, Aaron J. Moss, Rachel Hartman, Cheskie Rosenzweig, Richa Gautam, Jonathan Robinson, and Leib Litman. The Bots Ruining Social Science Are Not Bots at All. *Perspectives on Psychological Science*, 21(2): 127–137, March 2026. ISSN 1745-6916. doi:10.1177/17456916251404872. URL <https://doi.org/10.1177/17456916251404872>.
- Felipe M. Affonso. Detecting Vision-Enabled AI Respondents in Behavioral Research Through Cognitive Traps, November 2025. URL <https://papers.ssrn.com/abstract=5766848>.
- Marybec Griffin, Richard J. Martino, Caleb LoSchiavo, Camilla Comer-Carruthers, Kristen D. Krause, Christopher B. Stults, and Perry N. Halkitis. Ensuring survey research data integrity in the era of internet bots. *Quality & Quantity*, 56(4):2841–2852, August 2022. ISSN 1573-7845. doi:10.1007/s11135-021-01252-1. URL <https://doi.org/10.1007/s11135-021-01252-1>.
- Gemma Loebenberg, Melissa Oldham, Jamie Brown, Larisa Dinu, Susan Michie, Matt Field, Felix Greaves, and Claire Garnett. Bot or Not? Detecting and Managing Participant Deception When Conducting Digital Research Remotely: Case Study of a Randomized Controlled Trial. *Journal of Medical Internet Research*, 25(1):e46523, September 2023. doi:10.2196/46523. URL <https://www.jmir.org/2023/1/e46523>. Company: Journal of Medical Internet Research Distributor: Journal of Medical Internet Research Institution: Journal of Medical Internet Research Label: Journal of Medical Internet Research.
- Veniamin Veselovsky, Manoel Horta Ribeiro, and Robert West. Artificial Artificial Artificial Intelligence: Crowd Workers Widely Use Large Language Models for Text Production Tasks, June 2023. URL <http://arxiv.org/abs/2306.07899>. arXiv:2306.07899 [cs].
- Dapeng Zhang, Marina Katoh, and Weiping Pei. Detecting the Use of Generative AI in Crowdsourced Surveys: Implications for Data Integrity, October 2025. URL <http://arxiv.org/abs/2510.24594>. arXiv:2510.24594 [cs].
- Benjamin D. Douglas, Patrick J. Ewell, and Markus Brauer. Data quality in online human-subjects research: Comparisons between MTurk, Prolific, CloudResearch, Qualtrics, and SONA. *PLOS ONE*, 18(3):e0279720, March 2023. ISSN 1932-6203. doi:10.1371/journal.pone.0279720. URL <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0279720>.
- Daniel OConnell, Ashley Bautista, Clint Johnson, and Amanda Venta. Caution when Crowdsourcing: Prolific as a Superior Platform Compared with MTurk. *Graduate Student Journal of Psychology*, 25, December 2025. ISSN 2166-9066. doi:10.52214/gsjp.v25i1.14129. URL <https://journals.library.columbia.edu/index.php/gsjp/article/view/14129>.
- Margaret A. Webb and June P. Tangney. Too Good to Be True: Bots and Bad Data From Mechanical Turk. *Perspectives on Psychological Science*, 19(6):887–890, November 2024. ISSN 1745-6916. doi:10.1177/17456916221120027. URL <https://doi.org/10.1177/17456916221120027>.
- Douglas J. Ahler, Carolyn E. Roush, and Gaurav Sood. The micro-task market for lemons: data quality on Amazon’s Mechanical Turk. *Political Science Research and Methods*, 13(1):1–20, January 2025. ISSN 2049-8470, 2049-8489. doi:10.1017/psrm.2021.57. URL <https://www.cambridge.org/core/journals/political-science-research-and-methods/>

- article/microtask-market-for-lemons-data-quality-on-amazons-mechanical-turk/B379D8827575D81857C872BB5C40B660.
- Oriane A M Georgeac. A Crisis of Unverifiable Data: IP Analyses May Be Key to Ensuring Data Integrity in Online Survey Research, February 2026. URL https://osf.io/preprints/psyarxiv/r69hk_v1/.
- Christopher Rodriguez and Daniel M. Oppenheimer. Creating a Bot-tleneck for malicious AI: Psychological methods for bot detection. *Behavior Research Methods*, 56(6):6258–6275, September 2024. ISSN 1554-3528. doi:10.3758/s13428-024-02357-9. URL <https://doi.org/10.3758/s13428-024-02357-9>.
- Thomas J. Shaw, Cory J. Cascalheira, Emily C. Helminen, Cal D. Brisbin, Skyler D. Jackson, Melissa Simone, Tami P. Sullivan, Abigail W. Batchelder, and Jillian R. Scheer. Yes stormtrooper, these are the droids you are looking for: Identifying and preliminarily evaluating bot and fraud detection strategies in online psychological research. *Psychological Methods*, 30(6):1294–1310, 2025. ISSN 1939-1463. doi:10.1037/met0000724.
- Josielli Comachio, Adam Poulsen, Adeola Bamgboje-Ayodele, Aidan Tan, Julie Ayre, Rebecca Raeside, Rajshri Roy, and Edel O’Hagan. Identifying and counteracting fraudulent responses in online recruitment for health research: a scoping review. *BMJ Evidence-Based Medicine*, 30(3):173–182, June 2025. ISSN 2515-446X, 2515-4478. doi:10.1136/bmjebm-2024-113170. URL <https://ebm.bmj.com/content/30/3/173>.
- Brittney Goodrich, Marieke Fenton, Jerrod Penn, John Bovay, and Travis Mountain. Battling bots: Experiences and strategies to mitigate fraudulent responses in online surveys. *Applied Economic Perspectives and Policy*, 45(2): 762–784, 2023. ISSN 2040-5804. doi:10.1002/aep.13353. URL <https://onlinelibrary.wiley.com/doi/abs/10.1002/aep.13353>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/aep.13353>.
- Zhicheng Lin. Synthetic respondents and the illusion of human data: Rethinking online data collection in the age of generative AI, January 2026. URL https://osf.io/preprints/psyarxiv/6q9ps_v2/.
- Alexis Danielle Webster. *Exploring response patterns to open-ended items as indicators of data quality*. PhD thesis, University of British Columbia, 2023. URL <https://open.library.ubc.ca/soa/cIRcle/collections/ubctheses/24/items/1.0435718>.
- Anjana Karumathil and Ritu Tripathi. Combating Survey Bots in Online Research: An Integrative Literature Review of Insights and Strategies. *AIS Transactions on Human-Computer Interaction*, 17(1):80–109, March 2025. ISSN 1944-3900. doi:10.17705/1thci.00218. URL <https://aisel.aisnet.org/thci/vol17/iss1/3>.
- Christine Cuskley and Justin Sulik. The Burden for High-Quality Online Data Collection Lies With Researchers, Not Recruitment Platforms. *Perspectives on Psychological Science*, 19(6):891–899, November 2024. ISSN 1745-6916. doi:10.1177/17456916241242734. URL <https://doi.org/10.1177/17456916241242734>.
- Evgenia Christoforou, Gianluca Demartini, and Jahna Otterbacher. Generative AI in Crowdfunder for Web and Social Media Research: A Survey of Workers at Three Platforms. *Proceedings of the International AAAI Conference on Web and Social Media*, 18:2097–2103, May 2024. ISSN 2334-0770. doi:10.1609/icwsm.v18i1.31452. URL <https://ojs.aaai.org/index.php/ICWSM/article/view/31452>.
- Marek Muszyński. Attention checks and how to use them: Review and practical recommendations. *ASK: Research & Methods*, 32(1):3, January 2023. ISSN 1234-9224. doi:10.18061/ask.v32i1.0001. URL <https://openurl.ebsco.com/contentitem/doi:10.18061%2Fask.v32i1.0001?sid=ebsco:plink:crawler&id=ebsco:doi:10.18061%2Fask.v32i1.0001>.
- Hao Peng, Jingyun Zhang, Xiang Huang, Zhifeng Hao, Angsheng Li, Zhengtao Yu, and Philip S. Yu. Unsupervised Social Bot Detection via Structural Information Theory. *ACM Trans. Inf. Syst.*, 42(6):148:1–148:42, August 2024. ISSN 1046-8188. doi:10.1145/3660522. URL <https://dl.acm.org/doi/10.1145/3660522>.
- Benjamin Lebrun, Sharon Temtsin, Andrew Vonasch, and Christoph Bartneck. Detecting the corruption of online questionnaires by artificial intelligence. *Frontiers in Robotics and AI*, 10, February 2024b. ISSN 2296-9144. doi:10.3389/frobt.2023.1277635. URL <https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2023.1277635/full>.